Reg. No. | | | | | | | | | | | [MSCCH101]

FIRST SEMESTER M.Sc. (CHEMISTRY) EXAMINATION, DECEMBER 2015

INORGANIC CHEMISTRY

Time : 03 Hours **Maximum Marks:80**

Instructions:
1. Answer **all** Questions
2. Questions **1 to 4** carry **15** marks each.
3. Question No. **5** carries **20** marks.

1a. Explain the effects of hydrogen bonding. Comment on the use of IR spectroscopy to detect hydrogen bonding.

1b. Describe the concept of hybridization in chemical bonding. Explain any three structures associated with different hybridizations with examples.

1c. Why are the following statements false?

i. all groups contain both metals and non-metals

ii. in group I, reactivity decreases with increase in atomic number

iii. elements in a period become more metallic with increasing atomic number.

iv. atoms of elements in the same group have the same number of electrons.

iv. electronegativity of atoms of elements remain constant **5+5+5=15**

OR

1d. **i.** Define lattice energy

ii. explain Born-Haber cycle.

1e. Draw and label the molecular orbital energy level diagram for F_2 molecule, write its electronic configuration and calculate the bond order.

1f. Using molecular orbital theory, justify the non-existence of He_2 and Be_2 molecules.

5+5+5=15

2a. **i.** What is three centre-two-electron bonding?. Explain with an example.

ii. Explain the reactivity of diborane with, NH_3, Na and H_2O.

2b. Write a note on polymorphism of carbon and phosphorous.

SCHOOL OF ENGINEERING & TECHNOLOGY LIBRARY DATE JAIN UNIVERSITY-BANGALORE

2c. What are oxyacids of suphur? Explain with examples. **5+5+5=15**

OR

2d. Explain the following in detail.

i. minerals, ii. silica gel and iii. glass ceramics

2e. What are silicates? Explain the classification of silicates with examples.

2f. What are zeolites? Write their applications. **6+5+4=15**

3a. Explain conjugate acid-base pairs with examples.

3b. Write and explain the balanced equations for the dissociation of each of the following Bronsted- Lowry acids in water; H_2SO_4, HSO_4^-, H_3O^+ and NH_4^+.

3c. Brief on hard and soft acids and bases with examples.

3d. Comment on the use of BrF_3 as a non-aqueous solvent. **3+4+3+5=15**

OR

3e. List the dissimilarities between nuclear and chemical reactions.

3f. Prove that half-life of any radioactive substance is independent to the initial amount.

3g. Comment on the following:

i. bond order of O_2^{2-}.

ii. anhydrous H_2SO_4 as a non-aqueous solvent. **4+5+6=15**

4a. Draw the shape of a H_2S molecule and a H_3O^+ ion. Explain in terms of the valence shell electron pair repulsion theory.

4b. Write a note on sulphur-nitrogen compounds.

4c. i. Write the resonance structure of borazine.

ii. The concentration of OH^- in a sample of seawater is 5.0×10^{-6} mol·L^{-1}. Calculate the concentration of H_3O^+ ions, and classify the solution as acidic, basic or neutral.

5+5+5=15

OR

4d. Show by drawings and explain how *s-s*, *s-p*, *p-p* and *p-d* combination of atomic orbitals occur.

4e. What are condensed phosphates and polyhalides? Explain with examples.

4f. Explain radioactivity with an example. Calculate the half-life period of a nucleus if at the end of 4.2 days $N=0.798N_0$. **5+5+5=15**

5. **Answer any FOUR of the following.** **5+5+5+5+5=20**

a. Explain the following:

i. interhalogens and psuedohalogens.

ii. structure of XeF_2 and XeF_4

b. Discuss the formation of covalent bonds in a CO_3^{2-} ion using molecular orbital theory.

c. Explain the structure and bonding in diborane.

d. Define covalent, coordinate, multicentre quadruple and agnostic bonds.

e. Write a note on isomorphous replacements and condensed phosphates.

f. Explain liquid drop model.